Universal gravitational energy

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use the zero-at-infinity convention for gravitational potential energy.

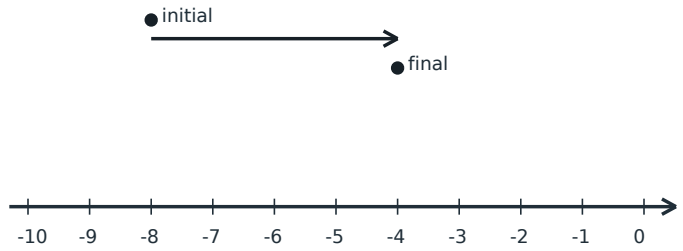
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 6.*
No. Work and kinetic-energy change must refer to the same endpoints. Measure matching intervals.

02. Worked example: explain the model
-40 to -20 moves toward zero; outward motion increases U.

03. Read and interpret the model
Less negative: U magnitude halves.

Model / diagram check



Example: potential energy (J)

04. Apply the physics
-8 J;-4 J;ΔU+4 J.

05. Explain your reasoning
Potential zero is convention; kinetic energy½mv² is nonnegative. Total can be negative for bound motion.

06. Review: connect an earlier idea *Review: Unit 3, Day 11.*
Shift becomes 4 times as large: shift=Δv×time. The simplified model omits orbital curvature/gravity and uncertainties.

07. Exit check
+4 J; gained.

08. STRETCH: test the model
-30 and -10 J; outward change +20 J.

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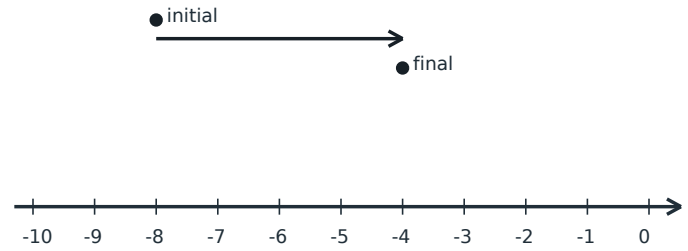
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 6.*
W=triangle area=1.5 J; v_f≈3.162 m/s.

02. Worked example: explain the model
-40 to -20 moves toward zero; outward motion increases U.

03. Read and interpret the model
Less negative: U magnitude halves.

Model / diagram check



Example: potential energy (J)

04. Apply the physics
U_i=-5.714×10¹⁰ J; U_f=-2.857×10¹⁰ J;ΔU=+2.857×10¹⁰ J.

05. Explain your reasoning
Potential zero is convention; kinetic energy½mv² is nonnegative. Total can be negative for bound motion.

06. Review: connect an earlier idea *Review: Unit 3, Day 11.*
Δv≈0.0006667 m/s; impulse≈5.333×10⁵ N·s. Uses constant added transverse velocity over a straight-line comparison.

07. Exit check
U_i=-12, U_f=-4;ΔU+8 J; gravity work -8 J.

08. STRETCH: test the model
ΔU=(2/3) GMm/R, gravity work negative of this; reaching infinity requires GMm/R total increase from R.